



Section: Our Dynamic Universe



Topic: Collisions, Explosions and Impulse

Question 3 – Train Wagon Collision

Given:

- Mass of moving wagon $m_1 = 7.5 \text{ tonnes} = 7500 \text{ kg}$
- Velocity of moving wagon $u_1 = 1.6 \text{ m/s}$
- Mass of stationary wagon $m_2 = 12.0 \text{ tonnes} = 12\,000 \text{ kg}$
- Initial velocity of second wagon $u_2 = 0$
- Wagons **join together** → inelastic collision

(a) Calculate the velocity of the wagons after the collision

✓ 3 marks

Use conservation of momentum:

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

$$7500 \times 1.6 + 12\,000 \times 0 = (7500 + 12\,000) v$$

$$12\,000 = 19\,500 v \Rightarrow v = \frac{12\,000}{19\,500} = 0.615 \text{ m/s}$$



Answer: 0.615 m/s

(b) State whether kinetic energy is conserved in this collision

✓ 1 mark



Answer:

No, kinetic energy is not conserved.

This is an **inelastic collision**, as the wagons **stick together**.

(c) Calculate the kinetic energy lost in the collision

✓ 3 marks

Initial KE:

$$\text{KE}_{\text{initial}} = \frac{1}{2} m_1 u_1^2 = \frac{1}{2} \times 7500 \times (1.6)^2 = 9600 \text{ J}$$

Final KE:

$$\text{KE}_{\text{final}} = \frac{1}{2} \times 19\,500 \times (0.615)^2 = 3692 \text{ J}$$

Loss in KE:

$$9600 - 3692 = 5908 \text{ J}$$



Answer: 5908 J lost



Revision Tips

- **Momentum** is always conserved in collisions
- **Kinetic energy** is only conserved in **elastic** collisions
- In **inelastic collisions**, objects stick together → shared final velocity